



Exploratory Data Analysis

Assessment 1 · Group030

FIT5196 Data Wrangling

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1. Context, preparation and analytical approach

This report examines commercial and operational patterns in six standardised relational tables created from the allocated Group030 JSON and XML sources: 5,000 orders, 15,723 order-item lines, 500 customers, 5,000 deliveries, 1,000 products and 7,000 reviews. It follows the EDA notebook sequence and recreates every reported statistic from the submitted CSV files.

The EDA begins with the six submitted relational tables rather than repeating the raw JSON/XML workflow. Four decisions from the solution are especially important here: source records were reconciled by normalised primary key before one canonical row was retained; item revenue and order totals were derived from reconciled values; narrative text was cleaned while the multilingual version and a separate Latin-analysis field were retained; and missing strings follow the required literal NaN contract. Examples are recorded in MAP-orders-10, MAP-orders-15, MAP-orders-22 and the corresponding review mappings.

The validation register provides the main assurance for analysis. All six primary keys are complete and unique (VAL-PK-01-VAL-PK-06), all eight submitted foreign-key relationships have zero orphan keys (VAL-FK-01-VAL-FK-08), and normalised cross-source overlaps contain no conflicting non-missing values (VAL-CONFLICT-01). Order-item revenue and order arithmetic reconcile with a maximum difference of \$0.00 (VAL-ARITH-00-VAL-ARITH-02). Delivery dates, delay fields and OTIF agree (VAL-TIME-01-VAL-TIME-03B), while the 303 reviews containing non-Latin script remain identified after cleaning (VAL-TEXT-02). These checks do not prove that every source value is correct, but they reduce the main risks of duplicate counting, broken joins and inconsistent derived measures.

Analytical approach

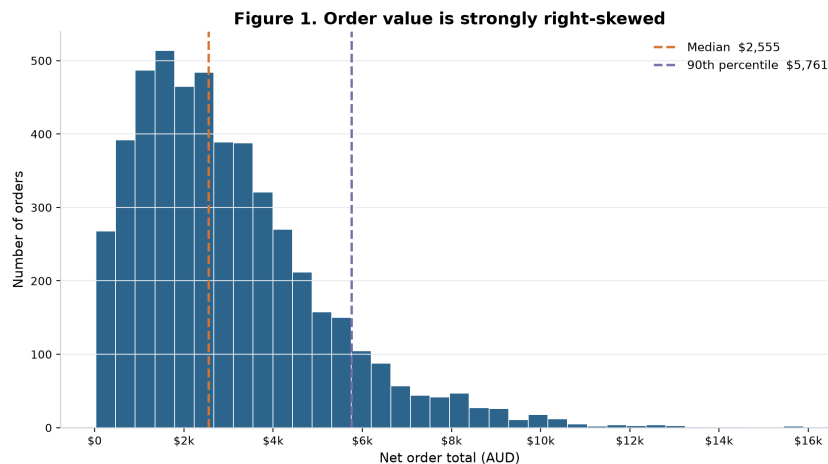
Identifiers are retained as strings, while the date, numeric and Boolean fields required for analysis are converted explicitly. Primary-key completeness and parent-key uniqueness are checked before each join.

Metrics are calculated at their stated observation unit. Order items join many-to-one to products; orders are aggregated before the customer join; and multiple reviews are averaged to order grain before joining deliveries. Every relational step confirms that the left-hand row count is unchanged.

The seven figures cover the six required EDA categories. They combine distributions, group comparisons, temporal rates, density, confidence intervals and checked relational analysis. The results describe associations in this export and are not interpreted as causal effects.

2. Assessed visualisations

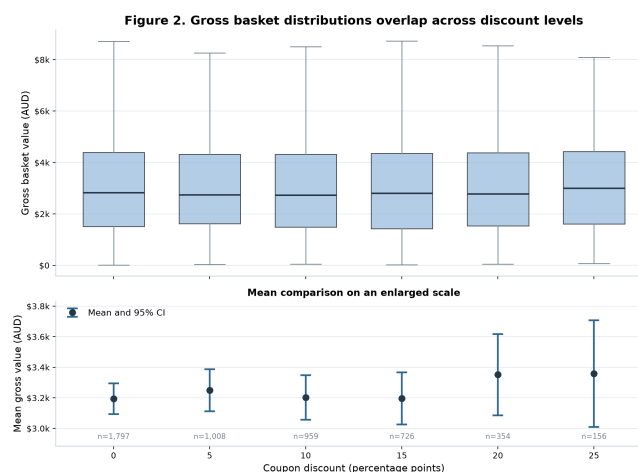
Figure 1. What does a typical order look like, and where does the high-value tail begin?



Unit and denominator: one order; all 5,000 canonical orders. **Table:** orders; no join. **Measure:** order_total, with the median and empirical 90th percentile.

The median order is \$2,555.44, while the highest 10% begin at \$5,760.76. The long upper tail means that a mean alone would overstate a typical order. This is a value threshold rather than a profitability threshold because costs and discounts are not allocated at order level.

Figure 2. Are deeper discounts associated with larger gross baskets?

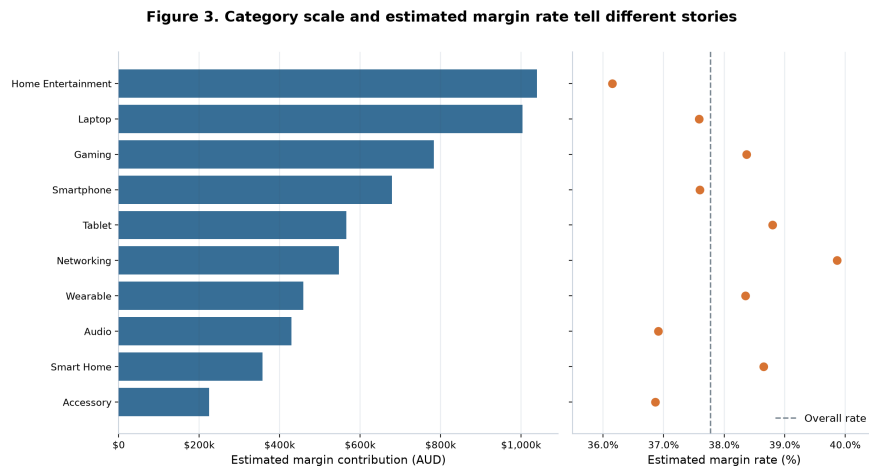


Unit and denominator: one order within each coupon-discount group; all 5,000 orders. **Table:** orders; no join. **Measure:** gross_order_price before discount, shown by median, interquartile range and mean.

Mean gross basket value ranges only from about \$3,195 to \$3,359, and the interquartile ranges overlap heavily. The 20% and 25% groups are much smaller, so their mean estimates are less precise. This is an observational comparison: coupon targeting and customer/product mix may differ between groups.

2. Assessed visualisations

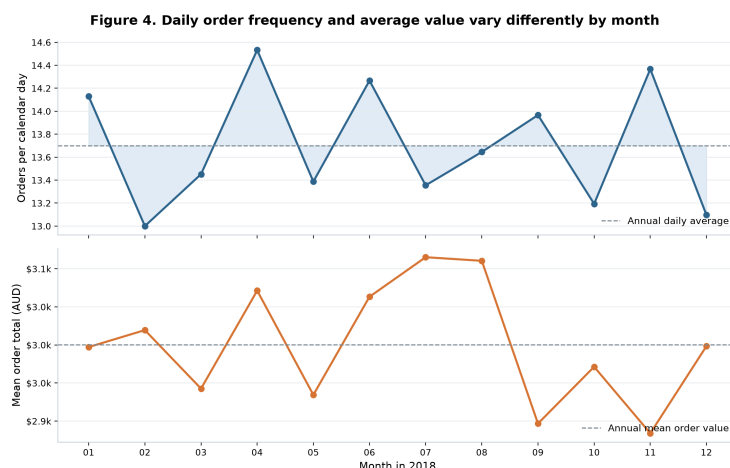
Figure 3. Which product categories combine economic scale with a strong estimated margin rate?



Unit and denominator: one canonical order-item line; all 15,723 lines. **Tables and join:** `order_items.product_id` → `products.product_id`, many-to-one. **Measures:** estimated margin contribution and estimated margin rate by category.

Home Entertainment and Laptop contribute the largest estimated margin dollars, while Networking has the highest estimated rate. These values use catalogue unit cost and do not allocate coupon discounts, returns or overhead, so they should be described as an exploratory contribution measure rather than accounting profit.

Figure 4. Do monthly changes come from order frequency or order value?

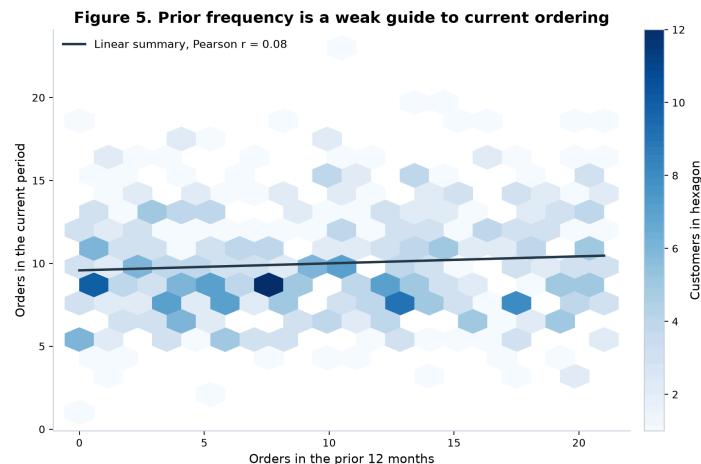


Unit and denominator: calendar month, calculated from all orders in that month. **Table:** `orders`; no join. **Measures:** orders per calendar day and mean net order value.

Daily order frequency and average order value move differently rather than following one common trend. With only one calendar year and no campaign calendar, the pattern is useful for description but not enough to claim seasonality.

2. Assessed visualisations

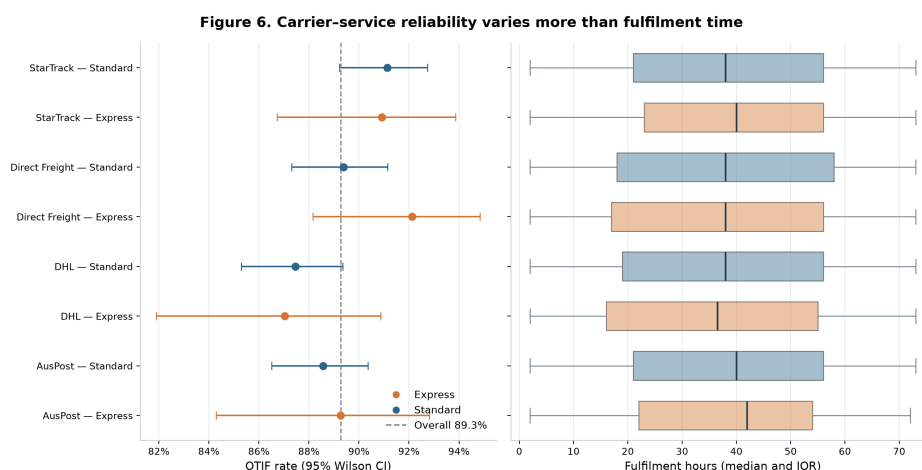
Figure 5. Does a customer's prior order frequency carry into the current period?



Unit and denominator: one customer; all 500 customers. **Tables and join:** current orders are first aggregated to customer grain, then joined to `customers.customer_id` one-to-one. **Measures:** prior 12-month orders and current-period order count.

The correlation is only **0.08** across 500 customers. Historical frequency alone therefore provides little separation in this period. This does not mean customer history is useless: tenure, marketing exposure and product mix may explain changes that a two-variable plot cannot show.

Figure 6. Which carrier and service combinations differ in reliability or fulfilment time?



Unit and denominator: one completed delivery within each carrier-service group; all 5,000 deliveries. **Tables and join:** the submitted delivery table already carries carrier, service, OTIF and fulfilment fields; its `order_id` relationship to orders was checked in the solution workflow. No extra metric-producing join is required here. **Measures:** OTIF rate with Wilson 95% CI, and fulfilment-hours distribution.

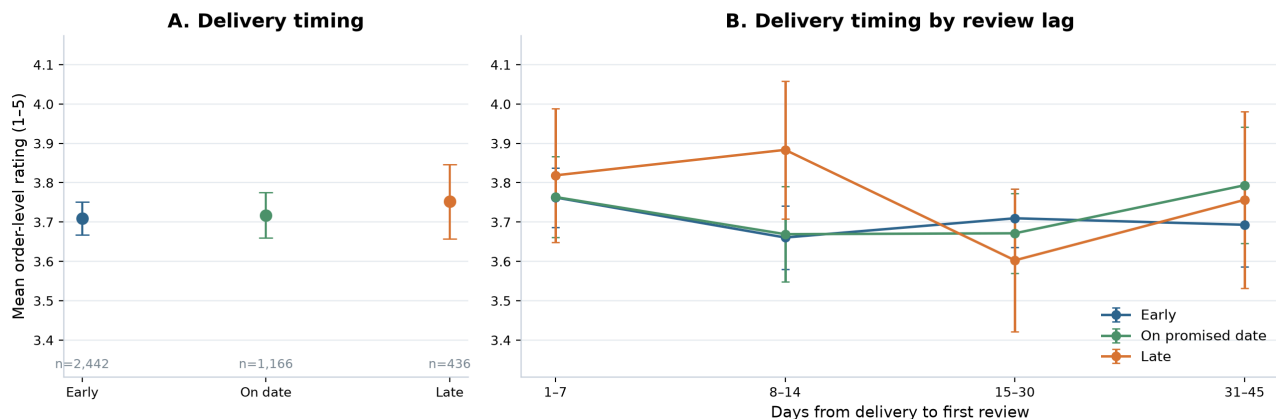
Overall OTIF is **89.3%**. The group estimates differ, but several intervals overlap, so the chart does not support a simple best-to-worst league table. Fulfilment-hour distributions also overlap substantially. Route mix, warehouse workload and destination distance remain plausible alternative explanations.

2. Assessed visualisations

Figure 7. Are delivery timing and review delay associated with the rating left for an order?

Figure 7. Rating differences are small across delivery and review timing

Enlarged vertical scale; all values remain within the original 1–5 rating scale.



Unit and denominator: one order with at least one review; 4,044 rated orders. Multiple reviews are averaged within order before joining. **Tables and join:** order-level `product_reviews.order_id` → `deliveries.order_id`, one-to-one after aggregation. **Measures:** mean order-level rating by early/on-date/late delivery and by days from delivery to first review, with 95% CIs on the full 1–5 scale.

Mean order-level ratings are **3.71 for early**, **3.72 for on-date** and **3.75 for late** deliveries. The confidence intervals overlap, and the review-lag panel has no stable monotonic pattern. This differs from the simple expectation that late delivery must produce a lower mean rating. Reviewer selection, product experience and the small late-delivery groups within each lag band could mask an association; this descriptive figure is not a causal test.

3. Evidence-based findings

- Order value is concentrated in a long upper tail (Figure 1).** Across all 5,000 canonical orders, the median net total is **\$2,555.44**, while the highest 10% begin at **\$5,760.76**. The tail matters because a small group of orders accounts for unusually high transaction value. Product mix and quantity may explain these orders, so the threshold is suitable for investigation rather than automatic customer treatment.
- Deeper discounts are not accompanied by a clear increase in gross basket size (Figure 2).** Mean gross order value ranges from **\$3,194.58** to **\$3,358.61** across the six discount groups, and their distributions overlap substantially. The 20% and 25% groups contain only 354 and 156 orders, so their confidence intervals are wider. Coupon allocation is observational; a controlled campaign would be needed before concluding that discount depth changes spending.

3. Evidence-based findings

- 3 **Home Entertainment and Laptop provide the largest estimated margin pools (Figure 3).** At order-item grain, Home Entertainment contributes approximately **\$1.040 million** and Laptop **\$1.004 million** across 15,723 lines. This suggests that changes in these categories could have a large aggregate effect. The calculation uses catalogue cost and excludes allocated coupons, returns and overhead, so it is an operating indicator rather than accounting profit.
- 4 **The category with the highest estimated margin rate is not the largest contributor (Figure 3).** Networking has the highest estimated rate at **39.9%**, but its estimated contribution is about **\$547,039**, roughly half that of Home Entertainment. Rate and scale therefore lead to different category priorities. Before changing assortment, realised discounts and category-specific operating costs should be included.
- 5 **Monthly order frequency and average value do not move together (Figure 4).** Daily order frequency ranges from **13.0 orders in February to 14.53 in April**, while mean order value ranges from **\$2,884.01 in November to \$3,115.23 in July**. Capacity and revenue planning should therefore treat transaction volume and basket value separately. One year of data and no campaign calendar are insufficient to identify a stable seasonal pattern.
- 6 **Prior order frequency is a weak stand-alone guide to current activity (Figure 5).** Across all 500 customers, prior-12-month orders and current-period orders have a Pearson correlation of only **0.08**. A retention rule based only on previous frequency would provide limited separation. Tenure, campaign exposure and product preference may account for changes, so these factors should be tested before using the relationship operationally.
- 7 **Most deliveries meet the OTIF definition, but the remaining failure share is still material (Figure 6).** Of 5,000 deliveries, **89.3%** are on time and in full, leaving approximately one in nine outside the target. This provides a clear operational outcome for further investigation. The aggregate rate hides differences in route difficulty, service mix and warehouse conditions, so interventions should not be assigned from the overall figure alone.
- 8 **Observed carrier–service OTIF rates differ, but several uncertainty intervals overlap (Figure 6).** DHL Express has the lowest observed rate at **87.0% (n=216)** and Direct Freight Express the highest at **92.1% (n=254)**. The overlap in Wilson intervals means the sample does not support a definitive league table. A fair comparison would adjust for distance, warehouse, order mix and repeated performance over time.
- 9 **Express service does not consistently produce shorter fulfilment inside the warehouse (Figure 6).** Median fulfilment ranges from **36.5 to 42.0 hours** across carrier–service groups, and the distributions overlap broadly. This suggests that dispatch preparation and external transport speed should be studied separately. Carrier selection may also be related to route and order characteristics, which are alternative explanations for the observed differences.
- 10 **Average rating shows no clear disadvantage for late delivery in this sample (Figure 7).** Among 4,044 rated orders, mean ratings are **3.71 for early (n=2,442)**, **3.72 for on-date (n=1,166)** and **3.75 for late delivery (n=436)**, with overlapping confidence intervals. This does not show that delays are harmless: only reviewed orders are observed, and product experience may dominate the average. Low-rating risk and delay severity deserve separate analysis.

4. Future machine-learning questions

MLQ-1

Question and decision	Can we identify orders at risk of OTIF failure early enough to prioritise intervention? Figure 6 shows a 10.7% overall failure share and uncertain carrier–service differences.
Unit and target	One order; binary target <code>on_time_in_full=False</code> .
Decision-time predictors	Warehouse, order time, basket value and size, expedited flag, carrier, service level and shipping distance, using only fields known by dispatch.
Validation	Forward time split; PR-AUC, calibration and recall at the intervention capacity.
Risk	Delivered date, delay reason and completed tracking events would leak the outcome; compare errors across locations and service levels.

MLQ-2

Question and decision	Can fulfilment hours be predicted when warehouse work begins so staffing or escalation can be planned? Figure 6 shows broad and overlapping fulfilment distributions.
Unit and target	One delivery/order; regression target <code>fulfilment_hours</code> .
Decision-time predictors	Order timestamp, warehouse, item count, total quantity, basket value, expedited flag, service level and distance available at fulfilment start.
Validation	Rolling temporal split; MAE and 90th-percentile absolute error.
Risk	Dispatch/delivery timestamps and later tracking events leak the target; monitor systematic underprediction by warehouse and service.

MLQ-3

Question and decision	Which completed orders are at risk of receiving a 1–2 star review so service recovery can be prioritised? Figure 7 shows that simple delivery timing means do not separate ratings well.
Unit and target	One completed order that is eligible for review; binary low-rating target after aggregating multiple reviews consistently.
Decision-time predictors	Product/category mix, customer history and delivery facts available before outreach.
Validation	Time-based split by order completion; PR-AUC, recall at contact capacity and calibration.
Risk	Review text, helpful votes and realised rating fields leak the outcome; non-reviewers create selection bias and performance should be checked by language.

4. Future machine-learning questions

MLQ-4

Question and decision	Can the number of orders a customer will place in the next 90 days support retention planning? Figure 5 shows prior frequency alone has weak association.
Unit and target	One customer at a fixed prediction date; count target of orders in the following 90 days.
Decision-time predictors	Orders strictly before the prediction date, recency/frequency/value summaries, tenure, tier, acquisition source and previous category mix.
Validation	Successive cohort-based temporal splits; MAE and Poisson deviance.
Risk	Future orders and retrospectively updated lifetime value leak the target; customer attributes may proxy sensitive characteristics.

MLQ-5

Question and decision	Can products be grouped into stable commercial roles for assortment review? Figure 3 shows that contribution and margin rate describe different category positions.
Unit and target	One product; unsupervised clusters based on lagged demand, revenue, estimated margin, price and review summaries.
Decision-time predictors	Only measures calculated from a completed historical window, standardised before clustering.
Validation	Compare stability across adjacent time windows; silhouette score plus merchant interpretability.
Risk	Scale choices can dominate clusters, sparse products may be unstable, and catalogue cost does not represent full realised profitability.

5. Limitations and conclusion

This analysis covers one retailer and one calendar year. It does not include campaign schedules, returns, realised operating costs, warehouse workload or route difficulty. Coupon, carrier and service assignments were not random, and product reviews represent customers who chose to leave feedback. The figures therefore describe associations in this export rather than causal effects. Estimated category margin should not be read as accounting profit, and the monthly analysis is too short to establish seasonality.

The strongest descriptive signals are the concentration of estimated category contribution and the remaining OTIF failure share. By contrast, discount depth does not clearly separate gross basket size, prior frequency is weak on its own, and average ratings differ little by delivery timing. These weak relationships are still useful because they show where simple rules would be unreliable. The next stage should use time-based evaluation, prevent outcome leakage and report uncertainty and subgroup performance before any operational use.

References

- NIST/SEMATECH. (n.d.). *Exploratory data analysis techniques*. NIST/SEMATECH e-Handbook of Statistical Methods. <https://www.itl.nist.gov/div898/handbook/eda/section3/eda3.htm> - Ravula, P. (2023). Impact of delivery performance on online review ratings: The role of temporal distance of ratings. *Journal of Marketing Analytics, 11*(2), 149–158. <https://doi.org/10.1057/s41270-022-00168-5> - Wikimedia Commons contributors. (2017). *Monash University logo* [SVG]. Wikimedia Commons. https://commons.wikimedia.org/wiki/File:Monash_University_logo.svg

The external study is used as motivation only. We do not reproduce its causal model or treat its reported effects as Group030 results.